Bihar Engineering University, Patna
VIIIth Semester Examination - 2023

Course: B.Tech. *Time: 03 Hours*
Code: 105811 *Subject: Real time systems* *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) In real-time operating system
(i) all processes have the same priority
(ii) a task must be serviced by its deadline period
(iii) process scheduling can be done only once
(iv) kernel is not required

(b) In rate monotonic scheduling
(i) shorter duration job has higher priority
(ii) longer duration job has higher priority
(iii) priority does not depend on the duration of the job
(iv) none of the above

(c) In which scheduling, certain amount of CPU time is allocated to each process?
(i) Earliest deadline first scheduling
(ii) Proportional share scheduling
(iii) Equal share scheduling
(iv) none of the above

(d) Hard real-time operating system has _____ jitter than a soft real-time operating system.
(i) less (ii) more (iii) equal (iv) None of the above

(e) Real time systems must have
(i) preemptive kernels (ii) non-preemptive kernels
(iii) preemptive kernels or non-preemptive kernels
(iv) neither preemptive nor non-preemptive kernels

(f) The ___ scheduling algorithm schedules periodic tasks using a static priority policy with preemption
(i) earliest deadline first (ii) rate monotonic
(iii) first cum first served (iv) priority

(g) Preemptive, priority based scheduling guarantees
(i) hard real time functionality (ii) soft real time functionality
(iii) protection of memory (iv) none of the above

(h) The amount of time required for the scheduling dispatcher to stop one process and start another is known as
(i) event latency (ii) interrupt latency
(iii) dispatch latency (iv) context switch

(i) VxWorks is centred around
(i) wind microkernel (ii) linux kernel (iii) unix kernel (iv) none of the above

(j) The problem of priority inversion can be solved by
(i) priority inheritance protocol (ii) priority inversion protocol
(iii) both (i) and (ii) (iv) None of the above

Q.2 (a) List different types of timing constraints that can occur in a real time system. **[7]**

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(b) Explain the important differences between 'hard', 'firm' and 'soft' real-time systems. [7]

Q.3 (a) What are the distinguishing characteristics of 'periodic' , 'aperiodic' and 'sporadic' real-time tasks? [7]

(b) A system contains five jobs. There are three resources X, Y and Z. The resource requirements of the jobs are listed below: [7]

J1 : (X; 2)
J2 : NONE
J3 : (Y;1)
J4 : (X; 3(Z; 1))
J5 ; (Y; 4(Z; 2))

The priority J_i is higher than the priority of J_j for $i<j$. What are the maximum blocking times of the jobs under the non-preemptable critical section protocol and under the priority-ceiling protocol?

Q.4 (a) What is understood by jitter associated with a periodic task? Mention techniques by which jitter can be overcome. [7]

(b) Distinguish between 'relative deadline' and ' absolute deadline' with the help of a suitable diagram. [7]

Q.5 (a) How are deadlocks, unbounded priority inversions and chain blocking prevented using Priority Ceiling Protocol (PCP)? [7]

(b) What is required to synchronize the clocks in a distributed real time system? Compare the advantages and disadvantages of centralized and the distributed clock synchronization. [7]

Q.6 (a) Explain the Earliest Deadline First (EDF) algorithm. [7]

(b) Why are algorithms which can satisfactorily schedule real-time tasks on multiprocessors not satisfactory to schedule real-time tasks on distributed systems? [7]

Q.7 (a) Why real time data cannot use TCP protocol? Give reasons. [7]

(b) Determine whether the following set of periodic real time tasks is schedulable on a uni-processor using RMA: [7]

Task	Start time (ms)	Processing time (ms)	Period (ms)	Deadline (ms)
T_1	20	25	150	100
T_2	40	7	40	40
T_3	60	10	60	50
T_4	25	10	30	20

Q.8 Consider a system that has five periodic taks, A,B,C,D and E, and three processors P1, P2 and P3. The periods of A,B and C are 2 and their execution times are equal to 1. The periods of D and E are 8 and their execution times are 6. The phase of every taks is 0, that is, the first job of the task is released at time 0. The relative deadline of every task is equal to its period. [7x2=14]

(a) Show that if the tasks are scheduled dynamically on three processors according to the LST algorithm, some jobs in the system cannot meet their deadlines.

(b) Find a feasible schedule of the five tasks on three processors.

Q.9 Write short notes on ***any two*** of the following: [7x2=14]

(a) Priority-ceiling portal
(b) Weighted fair-queueing servers
(c) EDF algorithm
(d) Embedded systems

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